

## ORIGINAL ARTICLE

# Transgenerational disruption of functional 5-HT<sub>1A</sub>R-induced connectivity in the adult mouse brain by traumatic stress in early life

F Razoux<sup>1,2</sup>, H Russig<sup>1</sup>, T Mueggler<sup>2,4</sup>, C Baltes<sup>2</sup>, K Dikaiou<sup>2</sup>, M Rudin<sup>2,3</sup> and IM Mansuy<sup>1</sup>

Traumatic stress in early life is a strong risk factor for psychiatric disorders that can affect individuals across several generations. Although the underlying mechanisms have been proposed to implicate serotonergic transmission in the brain, the neural circuits involved remain poorly delineated. Using pharmacological functional magnetic resonance imaging in mice, we demonstrate that traumatic stress in postnatal life alters 5-HT<sub>1A</sub> receptor-evoked local and global functions in both, the exposed animals and their progeny when adult. Disrupted functional connectivity is consistent across generations and match limbic circuits implicated in mood disorders, but also networks not previously linked to traumatic stress. These findings underscore the neurobiology and functional mapping of transgenerational effects of early life experiences.

*Molecular Psychiatry* (2017) **22**, 519–526; doi:10.1038/mp.2016.146; published online 27 September 2016

## INTRODUCTION

Traumatic stress in early life constitutes a serious risk factor for the development of psychiatric disorders, metabolic diseases and cancer.<sup>1</sup> It induces these pathologies not only in the exposed individuals but also often in their descendants.<sup>2</sup> In the brain, the effects of traumatic stress in early life may arise from the concomitant imbalance of neuroendocrine pathways and neuro-modulatory circuits such as serotonergic, glutamatergic and dopaminergic systems.<sup>3</sup> Recent studies in humans have identified epigenetic changes in the glucocorticoid system of Holocaust survivors and their offspring, providing initial evidence for intergenerational effect of trauma in human.<sup>4</sup> Behavioral alterations by early traumatic stress have been modeled in C57Bl/6 mice using unpredictable maternal separation combined with unpredictable maternal stress (MSUS) subjected from birth to postnatal day 14. When adult, MSUS animals (F1) have impaired sociability, depressive-like behaviors, risk-taking behaviors and deficient memory.<sup>5–7</sup> Further, the offspring (F2) obtained by breeding F1 MSUS males with naive control females or F1 MSUS females with naive control males, as well as the offspring of F2 animals (F3) have behavioral symptoms similar to F1 animals, despite not experiencing any stress themselves.<sup>5–8</sup> In F2 offspring, MSUS experienced by the F1 generation was observed to alter the level of serotonergic 1A receptor (5-HT<sub>1A</sub>R) binding in several brain areas.<sup>5</sup> Based on these alterations, we examined local brain activity and global functional networks using functional magnetic resonance imaging (fMRI) in mice. The suitability of fMRI to study the induced modulation of functional networks has been demonstrated previously.<sup>9–11</sup> We applied pharmacological stimulation using a 5-HT<sub>1A</sub>R agonist to determine whether and how

5-HT<sub>1A</sub>R functions and evoked connectivity are affected by MSUS across generations.

## MATERIALS AND METHODS

This study was conducted in strict adherence to the Swiss Law for Animal Protection, approved by the veterinary office of the canton Zurich, Switzerland. Experiments were carried out in male and female C57Bl6 adult mice (Elevage Janvier, Le Genest-Saint-Isle, France) bred in-house. Animals were housed under reverse light/dark cycle (light on 19:00–07:00) with standard rodent chow and tap water *ad libitum*. Unpredictable maternal separation combined with MSUS was carried out as previously described.<sup>6</sup> In brief, mouse pups (first generation, F1) were separated from their mother (F0) unpredictably for 3 h daily from postnatal day 1 to 14 (PND 1–14). Unpredictable maternal stress consisting of 20-min restraint stress or 5-min forced swimming was applied randomly any time during the 3 h separation. Corresponding control dams and litters were generated in parallel but not subjected to any manipulation except for a cage change once a week until weaning (PND21). Once weaned, pups were housed in groups of animals subjected to the same treatment (MSUS or control) but from different dams to avoid litter effects. To produce second-generation (F2) animals, control and MSUS F1 males were mated with naive C57Bl6 primiparous females before fMRI analyses (the offspring were generated from experimental males rather than females to prevent interference of maternal care). Males were removed shortly after mating and had no contact with gestating females or with their progeny. F2 pups from MSUS and control groups were weaned at PND21 and similarly to F1, reared in mixed social groups to avoid litter effects. F2 mice grew undisturbed except for a cage change once a week.

Functional MRI experiments were carried out in adult animals (7–10 months) as previously described.<sup>12</sup> In brief, the investigator was blind to the group allocation. Animals were picked randomly, anesthetized with isoflurane (Abbott, Cham, Switzerland), endotracheally intubated,

<sup>1</sup>Laboratory of Neuroepigenetics, University and ETH Zurich, Brain Research Institute, Center for Neuroscience Zürich, Zurich, Switzerland; <sup>2</sup>Institute for Biomedical Engineering, University and ETH Zurich, Zurich, Switzerland and <sup>3</sup>Institute of Pharmacology and Toxicology, University of Zurich, Center for Neuroscience Zürich, Zurich, Switzerland. Correspondence: Professor I Mansuy, Brain Research Institute, University of Zurich/Swiss Federal Institute of Technology, Winterthurerstrasse 190, Pharma Research & Early Development, Zurich 8057 Switzerland.  
E-mail: mansuy@hifo.uzh.ch

<sup>4</sup>Current address: F Hoffmann-La Roche AG, pRED, Pharma Research & Early Development, DTA Neuroscience, Grenzacherstrasse 124, Basel CH4070, Switzerland.

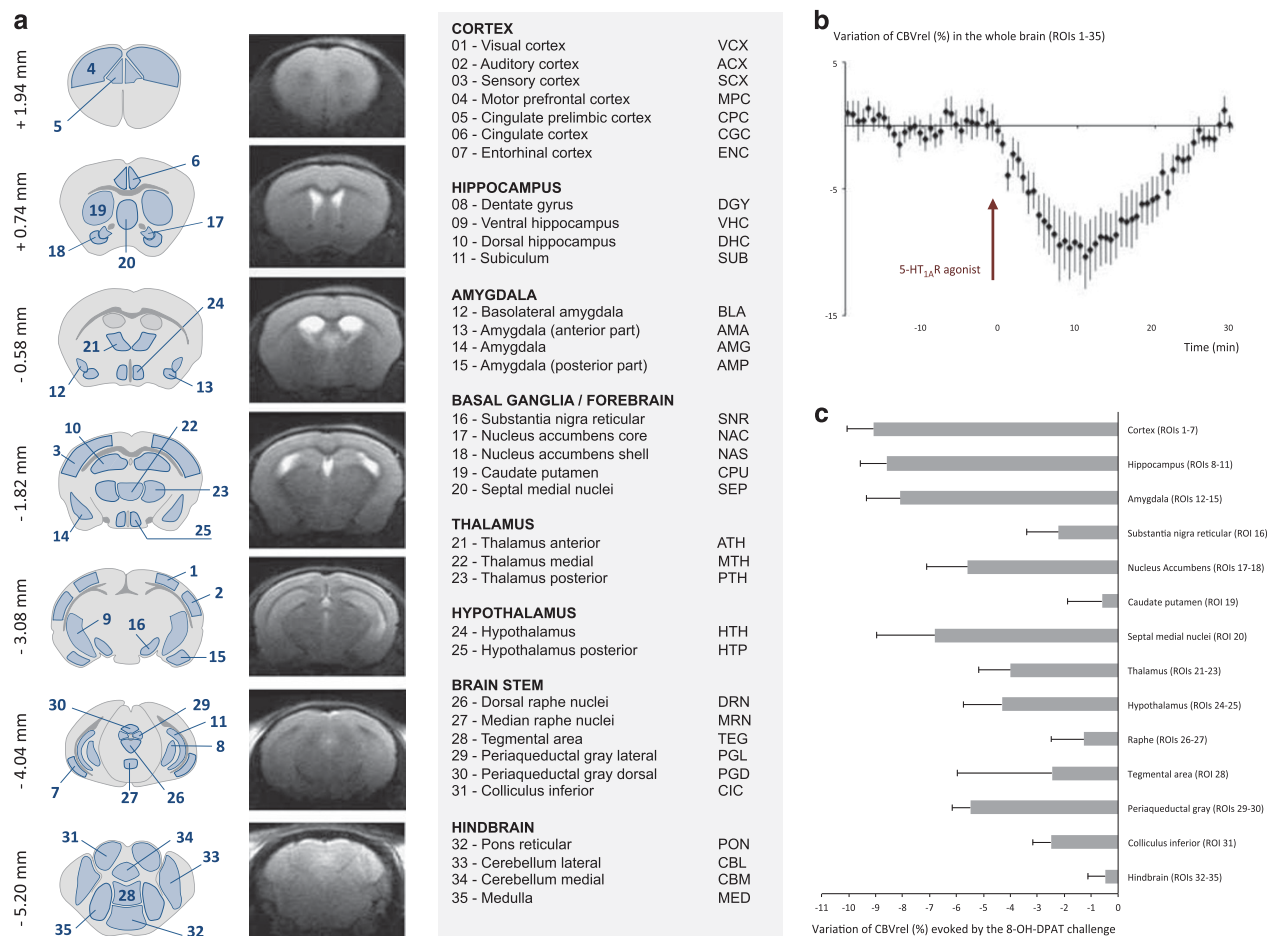
Received 11 December 2015; revised 6 April 2016; accepted 6 May 2016; published online 27 September 2016

artificially ventilated, and placed on a Plexiglas cradle with warm water circulation. Body temperature was maintained at  $36.5 \pm 0.5$  °C. The tail vein was cannulated for administration of the contrast agent and 8-OH-DPAT (0.1 mg kg<sup>-1</sup>). The animals received a single intravenous injection of gallamine triethiodide (15 mg kg<sup>-1</sup>) to minimize motion artifacts. Experiments were performed using a Pharmascan MR system 47/16 (field strength 4.7 Tesla, Bruker BioSpin, Ettlingen, Germany) equipped with a 200 MHz cryogenic transceiver radiofrequency probe to improve signal-to-noise ratio. Prior to fMRI, anatomical images were recorded using a multi-slice RARE spin echo sequence (eight slices with a spatial resolution of  $100 \times 100 \times 700$   $\mu\text{m}^3$  covering the whole brain). The fMRI protocol consisted of a RARE sequence with a spatial resolution of  $156 \times 156 \times 700$   $\mu\text{m}^3$ . Eight slices corresponding to the anatomical scan were recorded. Images were T<sub>2</sub>-weighted with a repetition delay of TR = 2500 ms and an echo delay of TE = 80.2 ms, and a RARE factor of 8, resulting in a temporal resolution of 40 s/multi-slice data set. Eight pre-contrast images ( $\bar{S}_{pre}$ ) were first acquired prior to administration of the contrast agent (Endorem, 55 mg Fe per kg). Eight-six sequential post-contrast images were acquired after a waiting period of 20-min later to allow the contrast agent to reach steady-state concentration in plasma. 8-OH-DPAT was administered after image 35. The average of images 5–35 yielded a baseline image ( $\bar{S}_0$ ) and subsequent images ( $S(t)$ ) reflected the effect of the drug at time  $t$ . In total, 35 regions of interest (ROIs) were defined across the brain including structures of the serotonergic pathways

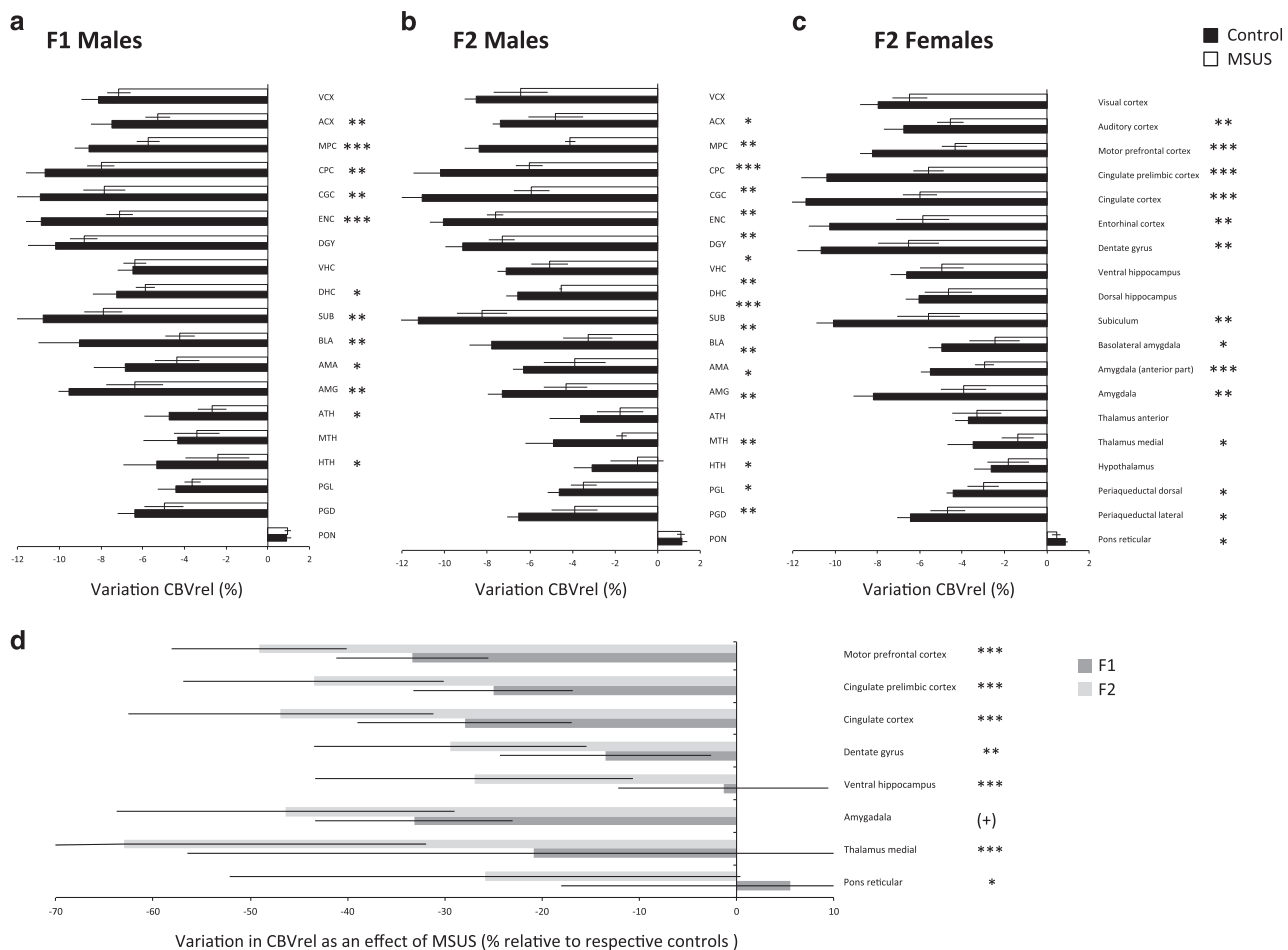
and control regions expressing no or low level of 5-HT<sub>1A</sub>R. All ROIs were manually drawn for each mouse and are listed in Figure 1. Delineation of ROIs and extraction of the corresponding MRI signal were carried out using Biomap (Novartis Institute for Biomedical Research, M Rausch). Prior to quantitative analyses of drug-induced cerebral blood volume (CBVrel, relative to baseline) response, data were de-trended to take the clearance of the contrast agent into account and corrected for MRI signal artifacts using Matlab (The Mathworks, Natick, MA, USA) as previously described.<sup>9</sup> De-trending was performed in each ROI individually by setting the five data points prior to drug administration and the last five data points of the serial data acquisition to  $0 \pm 5\%$ . Following de-trending, changes in CBVrel were computed for selected ROIs according to:

$$\Delta\text{CBV\%}(t) = \frac{\ln\{S(t)/\bar{S}_0\}}{\ln\{\bar{S}_0(t)/\bar{S}_{pre}\}} \cdot 100.$$

Functional networks elicited by pharmacological stimulation can be characterized by fMRI in rodents *in vivo*.<sup>10,11</sup> In a similar approach, we previously characterized the effect of genetic and pharmacological manipulations on the connectivity evoked by the activation 5-HT<sub>1A</sub>R in the mouse brain *in vivo*<sup>9</sup> and the present analyses of functional connectivity were carried out as previously described. Functional connectivity patterns derived from neuroimaging data were represented as networks, with anatomically defined structures being nodes, and measures of correlation between the response in each pair of nodes being edges. Cross-correlation (CC) was used as a measure of similarity of two signals, in



**Figure 1.** Local fMRI responses after serotonergic challenge with 8-OH-DPAT in the adult mouse brain *in vivo*. **(a)** Representative illustration of regions of interest (ROIs). Coronal sections of a mouse brain indicating the position of fMRI slices relative to Bregma (mm) (left) and corresponding representative fMRI images (middle). ROIs are represented in blue and are numbered and abbreviated for each brain structure (right). **(b)** Representative temporal profile of fMRI signal in a control mouse. The trace (mean effect and 95% confidence intervals) represents the variation in cerebral blood volume (CBVrel, % relative to baseline) in the whole brain (ROIs 1–35) before and after injection of the 5-HT<sub>1A</sub>R agonist 8-OH-DPAT. **(c)** Amplitude of variation in CBVrel (%) evoked by 8-OH-DPAT challenge (integrated CBV changes from 0 to 30 min relative to the respective baseline from –20 min to 0) in selected anatomical structures (all ROIs with homogenous responses) and selected ROIs in control mice (F1 and F2 males,  $n = 11$ ; results expressed as mean (s.d.), individual values for all ROIs are provided in Supplementary Table 1). CBV, cerebral blood volume; fMRI, functional magnetic resonance imaging.



**Figure 2.** MSUS treatment alters 5-HT<sub>1A</sub>-R-evoked fMRI responses across generations. Integrated  $\Delta$ CBVrel (%) changes (0–30 min) after 8-OH-DPAT injection in ROIs showing a significant difference in CBVrel response to the pharmacological challenge as an effect of MSUS treatment in (a) control and MSUS males from the first generation (F1 males, Control  $n=5$ , MSUS  $n=7$ ) (b) control and MSUS males from the second generation (F2 males, Control  $n=6$ , MSUS  $n=4$ ) and (c) control and MSUS females from the second generation (F2 females, Control  $n=4$ , MSUS  $n=6$ ) (FDR 5%  $*P < 0.05$ ,  $**P < 0.01$ ,  $***P < 0.001$ ). (d) Amplitude of variation in CBVrel in MSUS F1 (males,  $n=7$ ) and F2 (males and females,  $n=10$ ) animals relative to respective control mean in brain structures with significant changes as an effect of generation ( $*P < 0.05$ ,  $**P < 0.01$ ,  $***P < 0.001$  for F1 versus F2 animals). Results are expressed as mean (s.d.). CBV, cerebral blood volume; FDR, false discovery rate; fMRI, functional magnetic resonance imaging; MSUS, maternal stress; ROI, region of interest.

our case hemodynamic responses to drug stimulation in individual ROIs, as a function of a time-lag applied to one of them. Absolute coefficients of correlation were calculated according to:

$$r_{ij} = \frac{\sum_{k=0}^{N-1} (\Delta CBV_{\%i,j}(k \cdot \Delta t) - \overline{\Delta CBV_{\%i,j}}) \cdot (\Delta CBV_{\%j,i}(k \cdot \Delta t) - \overline{\Delta CBV_{\%j,i}})}{\sqrt{\sum_{k=0}^{N-1} (\Delta CBV_{\%i,j}(k \cdot \Delta t) - \overline{\Delta CBV_{\%i,j}})^2 \cdot \sum_{k=0}^{N-1} (\Delta CBV_{\%j,i}(k \cdot \Delta t) - \overline{\Delta CBV_{\%j,i}})^2}}$$

using Matlab (The Mathworks) where  $\Delta CBV_{\%i,j}(k \cdot \Delta t)$  represents the relative CBV change in ROIs  $i$  and  $j$ , respectively, at sampling point  $k$  of a data set comprising  $N$  sampling points ( $N=48$  in this study),  $\overline{\Delta CBV_{\%i,j}}$  the respective temporal average values, and  $\Delta t$  the sampling interval. Because the amplitude of responses was an important component of fMRI responses, correlation values were normalized to individual maximal amplitude. CC coefficients were calculated using  $r = r_{amp,ij} = r_{ij} \cdot \frac{\Delta CBV_{\%i,j}(\max) \cdot \Delta CBV_{\%j,i}(\max)}{(\max(\Delta CBV_{\%i,j}(\max), \Delta CBV_{\%j,i}(\max)))^2}$  for all ROI combinations individually for each animal. For representation purposes, individual computations were averaged across groups thereafter and the resulting correlation matrices ( $r$ ) were color-coded, with red indicating a high degree of similarity of fMRI response between brain regions, white no similarity and blue opposite similarity. The readers can contact the corresponding author for access to the MRI raw data and the Matlab code used to calculate the CC coefficients and generate the matrices.

Images and fMRI profiles with critical motion artifacts were excluded (two for F1 males, four for F2 males and five for F2 females) and imaging data were analyzed for F1 males (Control  $n=5$ , MSUS  $n=7$ ), F2 males (Control  $n=6$ , MSUS  $n=4$ ) and F2 females (Control  $n=4$ , MSUS  $n=6$ ). Effect size of 5-HT<sub>1A</sub>-R-evoked fMRI responses was evaluated by comparing mean  $\Delta$ CBV% values pre-injection (20 min of baseline) and post injection (0–30 min,  $t=0$  being the time point of drug administration) using one-way analysis of variance (ANOVA). The effect of MSUS on local fMRI response was tested by comparing mean  $\Delta$ CBV% value post injection using one-way ANOVA. The effect of generation on CBVrel was tested by comparing difference of individual mean  $\Delta$ CBV% values for MSUS animals relative to controls (group mean) using one-way ANOVA. The effect of 8-OH-DPAT administration on default connectivity between pairs of ROIs was tested by comparing CC coefficients of temporal profiles at baseline (20 min) and post injection (0–30 min) using one-way ANOVA. The effect of MSUS on connectivity between pairs of ROIs was tested by comparing CC coefficients for the post injection interval of the temporal profiles using one-way ANOVA. A false discovery rate (FDR) correction was applied using Mathworks BH procedure (post-hoc test that corrects for multiple comparisons, whose code can be found on <http://www.mathworks.com>). Owing to the high number of computations ( $n=1190$ ) used for the comparison of CC matrices, the FDR 5% correction applied in the analysis of connectivity is highly stringent (FDR 5%  $P < 0.05$  corresponds to an uncorrected threshold of uncorrected  $P < 0.0001$ ). A threshold of uncorrected  $P$ -value  $P < 0.01$  (corresponding to an FDR 32%  $P < 0.05$  in

F1 males and F2 females, and an FDR 27%  $P < 0.05$  in F2 males) was chosen to retain trends for connectivity changes. Results are expressed as mean (s.d.). Detailed CC coefficients,  $F$ , uncorrected  $p$  and FDR  $P$ -values are presented in Supplementary Information.

## RESULTS

Administration of the 5-HT<sub>1A</sub>R agonist 8-OH-DPAT to adult mice led to a relative region-specific decrease in CBVrel (Figure 1, Supplementary Table 1). Except for raphe nuclei, the largest CBVrel changes were observed in brain areas rich in 5-HT<sub>1A</sub>R such as limbic structures (hippocampus, amygdala and septum) and cortex (especially cingulate and entorhinal), whereas the smallest CBVrel changes were found in areas with low receptor density such as caudate putamen, substantia nigra and cerebellum. The response pattern evoked by 5-HT<sub>1A</sub>R activation was disrupted in F1 MSUS animals (Figures 2a, d; Supplementary Table 2). 8-OH-DPAT responses were significantly reduced in several brain structures including cortical areas, hippocampus, amygdala, thalamus and hypothalamus. fMRI analyses of the F2 offspring of F1 MSUS males showed disruption of 5-HT<sub>1A</sub>R-evoked responses in the same brain areas (Figures 2b–d; Supplementary Table 2). Response to 8-OH-DPAT was reduced in cortical areas, hippocampus, amygdala, thalamus and periaqueductal gray in both, male and female F2 offspring. The overall regional specificity of the effect was conserved across F1 and F2 generations, although loss of 5-HT<sub>1A</sub>R functions tended to amplify and extend to other brain structures in the F2 offspring (Figure 2d; Supplementary Table 2).

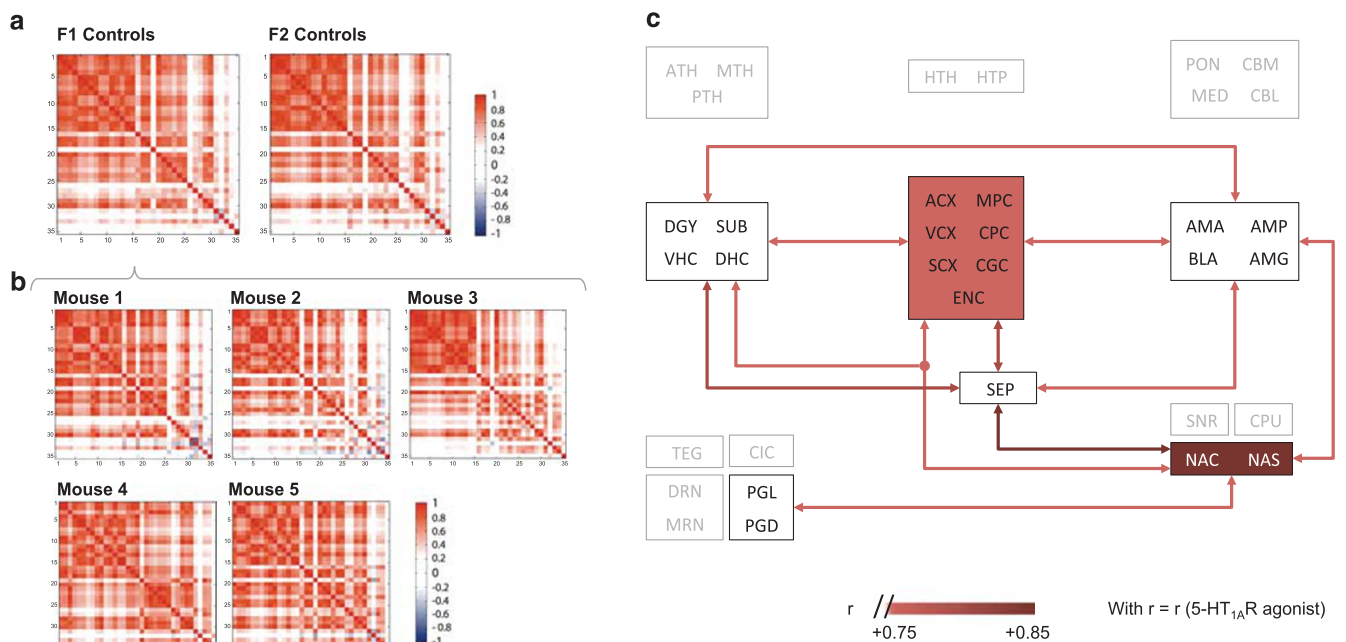
Administration of 5-HT<sub>1A</sub>R agonist to control males evoked functional connectivity responses characterized by high clustering with anatomical nodes such as cortex, hippocampus and amygdala (Figure 3, Supplementary Table 3). Overall, functional connectivity following 5-HT<sub>1A</sub>R challenge was highly reproducible across individuals and across generations (Figures 3a and b). The

strongest connectivity within and between structures was found in limbic structures including cortical areas, hippocampus, amygdala, septal nuclei and nucleus accumbens (Figure 3c).

Analysis of connectivity matrices in MSUS mice and their offspring revealed that 5HT<sub>1A</sub>R-evoked functional connectivity was compromised in both generations (Figure 4, Supplementary Figure 1, Supplementary Table 4). A major disruption of connectivity was detected between the amygdala (basolateral) and the hippocampus (subiculum) in the brain of F1 MSUS males. F2 MSUS males displayed reduced connectivity of the nucleus accumbens (core) with the amygdala and the visual cortex, reduced intra-cortical connectivity, and increased connectivity of the hippocampus (dentate gyrus) with the lateral cerebellum and motor prefrontal cortex.

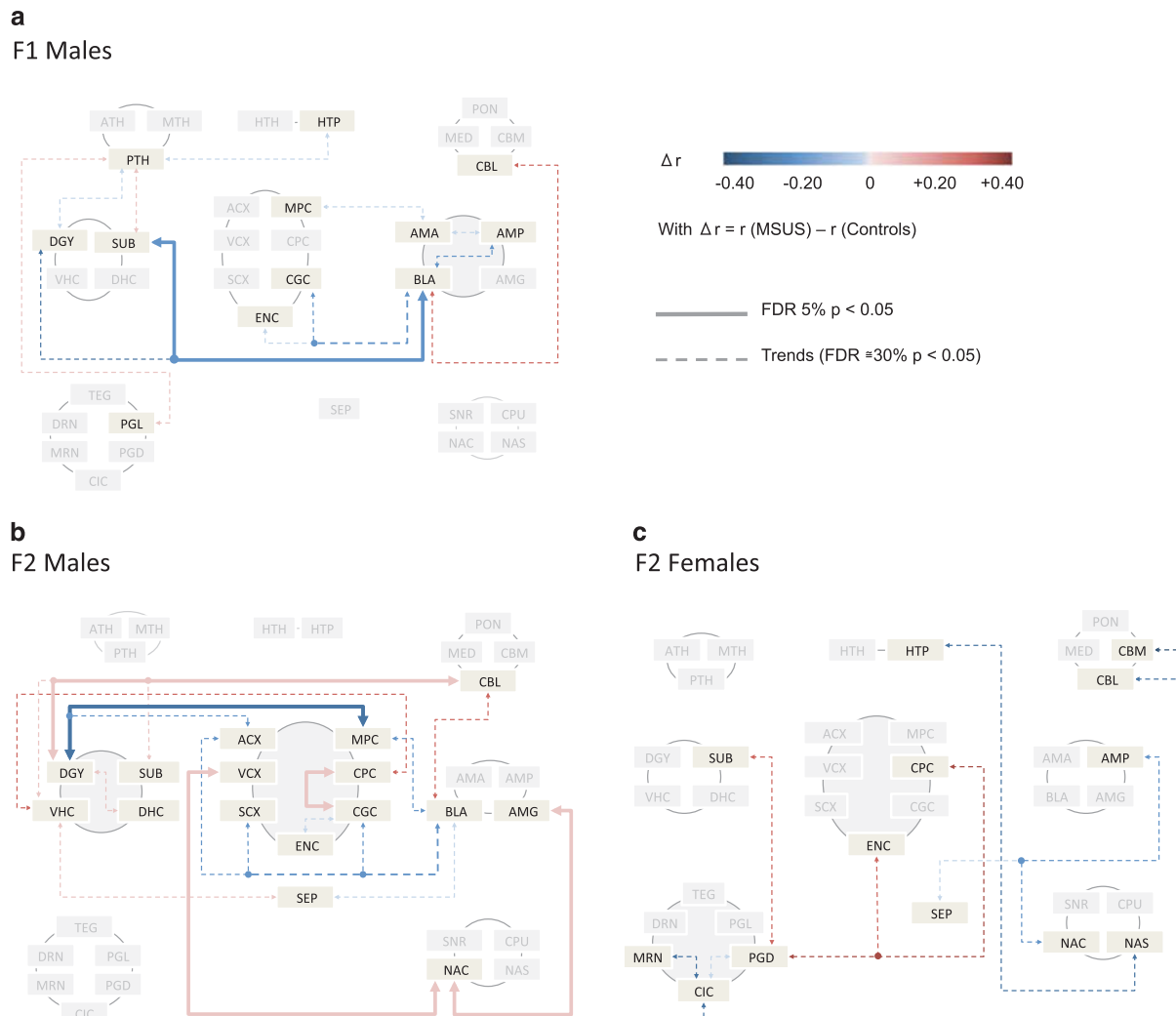
## DISCUSSION

This study demonstrates that serotonergic challenge by administration of 5-HT<sub>1A</sub>R agonist 8-OH-DPAT to adult mice exposed to traumatic stress in early postnatal life and their adult progeny strongly alters local cerebral blood volume (CBVrel, relative to baseline) response and functional connectivity throughout the brain. The 5-HT<sub>1A</sub>R is a receptor coupled to inhibitory G protein (G<sub>i</sub>/G<sub>o</sub>), whose activation leads to neuronal inhibition. Decreased neuronal activity results in a prolonged reduction in metabolic and hemodynamic demand that translates into lower CBVrel values. CBVrel changes induced by 8-OH-DPAT in adult mice reflect a decrease in 5HT<sub>1A</sub>R-evoked neuronal activity. This decrease was consistent across animals, and overall matched the distribution of 5-HT<sub>1A</sub>R.<sup>13</sup> The selectivity of the method was previously demonstrated using 5-HT<sub>1A</sub>R knockout animals.<sup>12,14</sup> Thus, CBVrel changes detected in regions expressing low level of 5-HT<sub>1A</sub>R such as the thalamus and nucleus accumbens<sup>13</sup> may reflect the contribution of additional neurotransmitter pathways connected to and/or interacting with the serotonergic system in



**Figure 3.** Global fMRI response to serotonergic challenge. (a and b) 5-HT<sub>1A</sub>R-evoked brain connectivity in control animals. Representative cross-correlation (CC) matrices in (a) control F1 ( $n = 5$ ) and F2 ( $n = 6$ ) males, and in (b) individual control mice (five different F1 males) after 8-OH-DPAT challenge. Color bar indicates value of CC coefficients  $r$  ( $r_{AMP,ij}$ ). (c) Graphical representation of evoked brain connectivity in the anatomical structures and ROIs of control animals selected in Figure 1. All ROIs are grouped by structure. Structures (mean of individual ROIs) and ROIs correspond to network nodes. Color-coded arrows indicate network edges with value of CC coefficients  $r$  ( $r_{AMP,ij}$ ) of the control group (F1 and F2 males,  $n = 11$ ). For purposes of representation, a threshold of  $r \geq 0.75$  was selected to enable the retention of ~15 connections. fMRI, functional magnetic resonance imaging; ROI, region of interest.





**Figure 4.** MSUS treatment alters serotonergic functional connectivity across generation. 5HT<sub>1A</sub>R-evoked functional connectivity in (a) F1 males (Control  $n=5$ , MSUS  $n=7$ ), (b) F2 males (Control  $n=6$ , MSUS  $n=4$ ) and (c) F2 females (Control  $n=4$ , MSUS  $n=6$ ). The network graphs show the differences in cross-correlation values  $r_{\text{amp},ij}$  ( $r$ ) observed between groups ( $\Delta r = r(\text{MSUS}) - r(\text{Control})$ ). Color-coded arrows indicate the amplitude of changes, with blue arrows indicating a reduction in  $r$  ( $\Delta r < 0$ ) and red, an increase ( $\Delta r > 0$ ) with respect to control. Plain lines represent significant differences retained after correction for multiple comparison (FDR 5%  $P < 0.05$ ) and dashed lines, connections not retained after correction but considered as trends (FDR 5% n.s., uncorrected  $P < 0.01$  corresponding to FDR ~ 30%  $P < 0.05$ ) (Supplementary Figure S1, Supplementary Table S4). FDR, false discovery rate; MSUS, maternal stress.

cortical structures, hippocampus or amygdala.<sup>15</sup> As previously shown,<sup>12</sup> the lack of significant CBVrel change in the dorsal raphe nucleus despite the presence of somatodendritic 5-HT<sub>1A</sub>R on the cell body of serotonergic neurons, most likely results from partial volume effects.

The disrupted response in F1 MSUS animals is consistent with altered serotonergic neurotransmission and 5-HT<sub>1A</sub>R mRNA expression reported in similar brain regions after early life stress in animal models (cf. Table 1).<sup>16–31</sup> Of particular interest is the decreased response in the hypothalamus and amygdala nuclei, which is consistent with altered 5-HT<sub>1A</sub>R binding in these structures. This alteration may underlie the perturbed development of the neuroendocrine system and dysfunctions of the hypothalamic-pituitary-adrenal axis in subjects exposed to early trauma, in patients with depression or post-traumatic stress disorder, and in animal models of these disorders respectively.<sup>3,32–34</sup> In addition to structures classically known to be implicated in early life stress and associated pathologies, our data newly reveal altered 5-HT<sub>1A</sub>R response in the auditory cortex,

highlighting the potential implication of auditory processing known to depend on the 5-HT<sub>1A</sub>R system,<sup>35</sup> in the impact of MSUS.<sup>36</sup>

Although the offspring of F1 MSUS males were not exposed to any traumatic stress, they showed similar functional impairments to F1 animals. Disruption of the 5HT<sub>1A</sub>R response detected in the hippocampus, thalamus and periaqueductal gray in F2 males is consistent with the decreased 5-HT<sub>1A</sub>R binding previously reported in these animals.<sup>5</sup> Likewise, the altered cortical response is in line with the increased level of serotonin in frontal cortex, and of a potentially decreased postsynaptic receptor sensitivity.<sup>1</sup> Given the importance of serotonergic neurotransmission for brain development and adult functions,<sup>37</sup> the persistence of an altered 5-HT<sub>1A</sub>R transmission likely underlies behavioral symptoms in the offspring.

In comparison with the methodology commonly used in resting-state fMRI studies, our method uses a lower temporal resolution (40 s versus ~1 s) which narrows down the range of signals that can be computed. This methodological difference

**Table 1.** Effect of ELS involving chronic maternal separation on 5-HT<sub>1A</sub>R expression and functions in adult monkey, mouse and rat

|               | CORTEX                                |                            | HIPPOCAMPUS                                                |                                                 | AMYGDALA                             |                                     | BASAL GANGLIA                |              | HYPOTHALAMUS                                              |                              | RAPHE                                              |                                            | REFERENCES                                                                                     |
|---------------|---------------------------------------|----------------------------|------------------------------------------------------------|-------------------------------------------------|--------------------------------------|-------------------------------------|------------------------------|--------------|-----------------------------------------------------------|------------------------------|----------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------|
|               | Area                                  | Effect                     | Area                                                       | Effect                                          | Area                                 | Effect                              | Area                         | Effect       | Area                                                      | Effect                       | Area                                               | Effect                                     |                                                                                                |
| <b>MOUSE</b>  | –                                     |                            | CA1<br>Dentate gyrus<br><b>CA1</b><br><b>Dentate gyrus</b> | n.s.<br>n.s.<br><b>(Reduced)</b><br><b>n.s.</b> | –                                    |                                     | –                            |              | –                                                         |                              | Dorsal<br>Median<br><b>Dorsal</b><br><b>Median</b> | n.s.<br>n.s.<br><b>n.s.</b><br><b>n.s.</b> | Advani <i>et al.</i> <sup>16</sup><br>Int. J. Neuropsychopharmacol.                            |
| <b>RAT</b>    | Prefrontal                            | Reduced                    | Dorsal                                                     | Reduced                                         | –                                    |                                     | –                            |              | –                                                         |                              | –                                                  |                                            | Shrestha <i>et al.</i> <sup>29</sup><br>Neurosci Lett.                                         |
|               | –                                     |                            | Dentate gyrus<br>CA1<br>CA3                                | n.s.<br>n.s.<br>n.s.                            | Basomedial<br>Basolateral<br>Central | Increased<br>Increased<br>Increased | –                            |              | –                                                         |                              | Dorsal                                             | Reduced                                    | Bravo <i>et al.</i> <sup>18</sup><br>Front Mol Neurosci.                                       |
|               | Prefrontal                            | Reduced                    | Whole structure                                            | Reduced                                         | –                                    |                                     | –                            |              | –                                                         |                              | –                                                  |                                            | Li <i>et al.</i> <sup>24</sup><br>Brain Res.                                                   |
|               | –                                     |                            | Whole structure                                            | Reduced                                         | –                                    |                                     | –                            |              | –                                                         |                              | –                                                  |                                            | Maniam & Morris <sup>25</sup><br>Psychoneuroendocrinology                                      |
|               | –                                     |                            | Ventral                                                    | n.s.                                            | –                                    |                                     | –                            |              | Unspecified                                               | Increased                    | Dorsal<br>Dorsal<br>Dorsal                         | n.s.<br>n.s.<br>n.s.                       | Lambas-Senas <i>et al.</i> <sup>20</sup><br>Prog. in NPP & Biol Psychiatry                     |
|               | –                                     |                            | –                                                          |                                                 | –                                    |                                     | –                            |              | –                                                         |                              | –                                                  |                                            | Oreland <i>et al.</i> <sup>27</sup><br>Brain Res.                                              |
|               | Cingulate<br>Prefrontal<br>Entorhinal | Reduced<br>Reduced<br>n.s. | Ventral CA1<br>Dorsal CA1<br>CA2<br>CA3                    | Reduced<br>n.s.<br>n.s.<br>n.s.                 | Basolateral<br>Medial<br>Caudal      | n.s.<br>n.s.<br>n.s.                | Accumbens<br>Dorsal striatum | n.s.<br>n.s. | Paraventricular<br>Ventromedial                           | n.s.<br>n.s.                 | Dorsal                                             | Reduced                                    | Leventopoulos <i>et al.</i> <sup>23</sup><br>Neuropharmacology                                 |
|               | Frontal<br>Parietal<br>Entorhinal     | n.s.<br>n.s.<br>n.s.       | Dentate gyrus<br>CA1<br>CA2<br>CA3                         | n.s.<br>n.s.<br>n.s.                            | Basolateral<br>Basomedial            | n.s.<br>n.s.                        | Accumbens                    | n.s.         | Paraventricular<br>Ventromedial<br>Dorsomedial<br>Lateral | n.s.<br>n.s.<br>n.s.<br>n.s. | –                                                  |                                            | Vicentic <i>et al.</i> <sup>31</sup><br>Neuroscience                                           |
|               | –                                     |                            | –                                                          |                                                 | –                                    |                                     | –                            |              | –                                                         |                              | Dorsal<br>Dorsal<br>Dorsal                         | n.s.<br>n.s.<br><b>Increased</b>           | Arborelius <i>et al.</i> <sup>17</sup><br>Psychopharmacology (Berl)                            |
|               | –                                     |                            | –                                                          |                                                 | –                                    |                                     | –                            |              | –                                                         |                              | Dorsal                                             | n.s.                                       | Gartside <i>et al.</i> <sup>19</sup><br>Eur. J. Neurosci.                                      |
|               | –                                     |                            | Dorsal CA1<br>Dentate gyrus                                | n.s.<br>n.s.                                    | –                                    |                                     | Striatum                     | n.s.         | –                                                         |                              | Dorsal                                             | n.s.                                       | Neumaier <i>et al.</i> <sup>26</sup><br>Biol Psychiatry                                        |
| <b>MONKEY</b> | Whole structure                       | n.s.                       | Whole structure                                            | n.s.                                            | –                                    |                                     | –                            |              | –                                                         |                              | Whole structure                                    | (Increased)                                | Shrestha <i>et al.</i> <sup>28</sup><br>Am J Psychiatry                                        |
|               | Cingulate<br>Prefrontal               | Reduced<br>n.s.            | Whole structure<br><b>Whole structure</b>                  | n.s.<br><b>Reduced</b>                          | Whole structure                      | n.s.                                | –                            |              | –                                                         |                              | Whole structure                                    | n.s.                                       | Spinelli <i>et al.</i> <sup>30</sup><br>Biol Psychiatry                                        |
|               | Cingulate<br>Cingulate                | Reorganized<br>n.s.        | CA1<br>CA3<br>Dentate gyrus<br>Subiculum                   | Reduced<br>n.s.<br>n.s.<br>n.s.                 | Basal                                | n.s.                                | –                            |              | –                                                         |                              | –                                                  |                                            | Law <i>et al.</i> <sup>21,22</sup><br>Int. J. Neuropsychopharmacol.<br>Neuropsychopharmacology |

Abbreviations: ELS, early-life stress; n.s., non-statistically significant. Gray: Males and females. Light gray: Males only. Regular text represents changes in the level of receptor binding compared to non-ELS control animals, italic text represents changes in 5-HT<sub>1A</sub>R mRNA level, and bold text represents changes in functional correlates.

limits the connectivity that can be observed during baseline and after saline administration.<sup>9</sup> Thus, pharmacologically evoked connectivity readouts are not comparable with resting-state fMRI data but instead, represent unique means to probe the circuitry associated with alterations in specific transmitter turnover or receptor level.<sup>10,11</sup> The functional networks evoked by 5-HT<sub>1A</sub>R agonist in control males are consistent with previous observations.<sup>9</sup> The connectivity pattern evoked by the administration of 8-OH-DPAT may reflect functional interactions arising from anatomical connections between structures expressing high level of 5-HT<sub>1A</sub>R such as raphe nuclei, and structures expressing low level such as the nucleus accumbens and periaqueductal gray.<sup>13,38</sup> The functional heterogeneity of the structures in the activated circuits illustrates the complexity of the serotonergic system and its interactions with other neurotransmitter systems. Despite such heterogeneity, the reproducibility of the pattern of connectivity underscores the power of the present method to study the effects of traumatic stress.

The compromised connectivity in MSUS mice and their offspring suggests reduced efficiency of transfer of 5-HT<sub>1A</sub>R-dependent information in the affected circuits.<sup>39</sup> Interestingly, the impaired connectivity observed in MSUS mice corroborates the changes in 5-HT<sub>1A</sub>R-dependent binding, impaired neuropsychological performance and disrupted connectivity reported in the amygdala and the cingulate cortex of human subjects exposed to chronic stress.<sup>40</sup> Connectivity changes in the amygdala have also been associated with an increased vulnerability to develop affective disorders following exposure to early life stress.<sup>32,41</sup> Whether there is a more causal relationship between these resting-state fMRI findings and altered 5-HT<sub>1A</sub>R-binding potential needs to be examined. Notably, the networks disrupted in MSUS animals from both generations substantially overlap with circuits known to be involved in behavioral responses such as social behavior, behavioral control and cognitive functions.<sup>5–8,42,43</sup> Thus, our results suggest that a failure to maintain 5-HT<sub>1A</sub>R-related connectivity following trauma exposure may critically contribute to stress vulnerability later in life in exposed individuals but also in their progeny.

The more extended changes in 5-HT<sub>1A</sub>R-evoked connectivity between the first and second generation of animals may result from the fact that F2 animals carry the alterations starting at fertilization and throughout development since germ cells have been implicated in transmission,<sup>6,42</sup> whereas F1 males are exposed to MSUS only after birth. This implies that neural connections in F2 animals are likely affected earlier than in F1 animals, therefore potentially more broadly and severely. Alterations in 5-HT<sub>1A</sub>R connectivity in F2 MSUS animals are also more pronounced than in heterozygous mice deficient for 5-HT<sub>1A</sub>R,<sup>9</sup> suggesting that brain systems other than depending on 5-HT<sub>1A</sub>R may also be affected. Such composite alterations may underlie some of the complexity of behavioral symptoms in MSUS animals across generations.<sup>5–8,42,43</sup>

Disruption of 5-HT<sub>1A</sub>R-evoked connectivity in F2 MSUS mice is gender-specific, a characteristic also observed in the global reorganization of functional networks after childhood maltreatment in human.<sup>44</sup> Altered 5-HT<sub>1A</sub>R-dependent connectivity in male MSUS offspring may contribute to the reduced sociability of the animals<sup>5</sup> as social interactions depend on hippocampal 5-HT<sub>1A</sub>R functions.<sup>45</sup> The connectivity between the amygdala and cingulate cortex is compromised in social anxiety and autism spectrum disorders,<sup>46,47</sup> disorders also associated with functional impairments of the 5-HT<sub>1A</sub>R. Whether alterations in social behaviors in F2 male mice may further arise from a reduced 5-HT<sub>1A</sub>R-dependent connectivity of this axis still needs to be demonstrated.

Although the changes in local CBVrel will ultimately impact functional connectivity, the precise relationship between local changes in CBVrel and global changes in functional connectivity was not demonstrated so far. In line with the accumulating

evidence that behavioral endophenotypes relate more closely to functional networks than to local brain activity,<sup>48</sup> we can note that the heterogeneity of the behavioral phenotypes exhibited by MSUS animals seem to relate more closely to specific changes in functional connectivity than to an overall reduction in CBVrel across structures.

In conclusion, this study demonstrates that early trauma alters 5-HT<sub>1A</sub>R-evoked functional connectivity in the adult mouse brain across generations. Functional impairments affect brain regions and circuits known to be involved in the behavioral symptoms observed in the animals. The data provide novel insight into the neurobiology of stress-induced transgenerational risk for psychiatric disorders. If translatable to humans, the identified alterations in connectivity represent promising novel targets for diagnostics and the design of potential therapeutic approaches.

## CONFLICT OF INTEREST

The authors declare no conflict of interest.

## ACKNOWLEDGMENTS

This study was supported by the University of Zurich, the Swiss Federal Institute of Technology, the Swiss National Science Foundation, the National Center of Competence in Research 'Neural Plasticity and Repair' and Roche. We thank Ry Tweedie-Cullen and Lukas von Ziegler for help with statistics.

## REFERENCES

- 1 Ehler U. Enduring psychobiological effects of childhood adversity. *Psychoneuroendocrinology* 2013; **38**: 1850–1857.
- 2 Bowers ME, Yehuda R. Intergenerational transmission of stress in humans. *Neuropsychopharmacology* 2016; **41**: 232–244.
- 3 Franklin TB, Saab BJ, Mansuy IM. Neural mechanisms of stress resilience and vulnerability. *Neuron* 2012; **75**: 747–761.
- 4 Yehuda R, Daskalakis NP, Bierer LM, Bader HN, Klengel T, Holsboer F *et al*. Holocaust exposure induced intergenerational effects on FKBP5 methylation. *Biol Psychiatry* 2015; **80**: 372–380.
- 5 Franklin TB, Linder N, Russig H, Thony B, Mansuy IM. Influence of early stress on social abilities and serotonergic functions across generations in mice. *PLoS One* 2011; **6**: e21842.
- 6 Franklin TB, Russig H, Weiss IC, Graff J, Linder N, Michalon A *et al*. Epigenetic transmission of the impact of early stress across generations. *Biol Psychiatry* 2010; **68**: 408–415.
- 7 Bohacek J, Farinelli M, Mirante O, Steiner G, Gapp K, Coiret G *et al*. Pathological brain plasticity and cognition in the offspring of males subjected to postnatal traumatic stress. *Mol Psychiatry* 2015; **20**: 621–631.
- 8 Weiss IC, Franklin TB, Vizi S, Mansuy IM. Inheritable effect of unpredictable maternal separation on behavioral responses in mice. *Front Behav Neurosci* 2011; **5**: 3.
- 9 Razoux F, Baltes C, Mueggler T, Seuwen A, Russig H, Mansuy I *et al*. Functional MRI to assess alterations of functional networks in response to pharmacological or genetic manipulations of the serotonergic system in mice. *Neuroimage* 2013; **74**: 326–336.
- 10 Schwarz AJ, Gozzi A, Bifone A. Community structure in networks of functional connectivity: resolving functional organization in the rat brain with pharmacological MRI. *Neuroimage* 2009; **47**: 302–311.
- 11 Schwarz AJ, Gozzi A, Reese T, Bifone A. In vivo mapping of functional connectivity in neurotransmitter systems using pharmacological MRI. *Neuroimage* 2007; **34**: 1627–1636.
- 12 Mueggler T, Razoux F, Russig H, Buehler A, Franklin TB, Baltes C *et al*. Mapping of CBV changes in 5-HT(1A) terminal fields by functional MRI in the mouse brain. *Eur Neuropsychopharmacol* 2011; **21**: 344–353.
- 13 Hannon J, Hoyer D. Molecular biology of 5-HT receptors. *Behav Brain Res* 2008; **195**: 198–213.
- 14 Gozzi A, Jain A, Giovannelli A, Bertolini C, Crestan V, Schwarz AJ *et al*. A neural switch for active and passive fear. *Neuron* 2010; **67**: 656–666.
- 15 Haber SN, Knutson B. The reward circuit: linking primate anatomy and human imaging. *Neuropsychopharmacology* 2010; **35**: 4–26.
- 16 Advani T, Hensler JG, Koek W. Effect of early rearing conditions on alcohol drinking and 5-HT1A receptor function in C57BL/6 J mice. *Int J Neuropsychopharmacol* 2007; **10**: 595–607.

- 17 Arborelius L, Hawks BW, Owens MJ, Plotsky PM, Nemeroff CB. Increased responsiveness of presumed 5-HT cells to citalopram in adult rats subjected to prolonged maternal separation relative to brief separation. *Psychopharmacology (Berl)* 2004; **176**: 248–255.
- 18 Bravo JA, Dinan TG, Cryan JF. Early-life stress induces persistent alterations in 5-HT<sub>1A</sub> receptor and serotonin transporter mRNA expression in the adult rat brain. *Front Mol Neurosci* 2014; **7**: 24.
- 19 Gartside SE, Johnson DA, Leitch MM, Troakes C, Ingram CD. Early life adversity programs changes in central 5-HT neuronal function in adulthood. *Eur J Neurosci* 2003; **17**: 2401–2408.
- 20 Lambas-Senas L, Mnie-Filali O, Certin V, Faure C, Lemoine L, Zimmer L et al. Functional correlates for 5-HT<sub>1A</sub> receptors in maternally deprived rats displaying anxiety and depression-like behaviors. *Prog Neuropsychopharmacol Biol Psychiatry* 2009; **33**: 262–268.
- 21 Law AJ, Pei Q, Feldon J, Pryce CR, Harrison PJ. Gene expression in the anterior cingulate cortex and amygdala of adolescent marmoset monkeys following parental separations in infancy. *Int J Neuropsychopharmacol* 2009; **12**: 761–772.
- 22 Law AJ, Pei Q, Walker M, Gordon-Andrews H, Weickert CS, Feldon J et al. Early parental deprivation in the marmoset monkey produces long-term changes in hippocampal expression of genes involved in synaptic plasticity and implicated in mood disorder. *Neuropsychopharmacology* 2009; **34**: 1381–1394.
- 23 Leventopoulos M, Russig H, Feldon J, Pryce CR, Opacka-Juffry J. Early deprivation leads to long-term reductions in motivation for reward and 5-HT<sub>1A</sub> binding and both effects are reversed by fluoxetine. *Neuropharmacology* 2009; **56**: 692–701.
- 24 Li M, Xue X, Shao S, Shao F, Wang W. Cognitive, emotional and neurochemical effects of repeated maternal separation in adolescent rats. *Brain Res* 2013; **1518**: 82–90.
- 25 Maniam J, Morris MJ. Voluntary exercise and palatable high-fat diet both improve behavioural profile and stress responses in male rats exposed to early life stress: role of hippocampus. *Psychoneuroendocrinology* 2010; **35**: 1553–1564.
- 26 Neumaier JF, Edwards E, Plotsky PM. 5-HT<sub>1B</sub> mRNA regulation in two animal models of altered stress reactivity. *Biol Psychiatry* 2002; **51**: 902–908.
- 27 Orelund S, Pickering C, Gokturk C, Orelund L, Arborelius L, Nylander I. Two repeated maternal separation procedures differentially affect brain 5-hydroxy-tryptamine transporter and receptors in young and adult male and female rats. *Brain Res* 2009; **1305**Suppl:537–49.
- 28 Shrestha SS, Nelson EE, Liow JS, Gladding R, Lyoo CH, Noble PL et al. Fluoxetine administered to juvenile monkeys: effects on the serotonin transporter and behavior. *Am J Psychiatry* 2014; **171**: 323–331.
- 29 Shrestha SS, Pine DS, Luckenbaugh DA, Varnas K, Henter ID, Innis RB et al. Anti-depressant effects on serotonin 1A/1B receptors in the rat brain using a gene x environment model. *Neurosci Lett* 2014; **559**: 163–168.
- 30 Spinelli S, Chefer S, Carson RE, Jagoda E, Lang L, Heilig M et al. Effects of early-life stress on serotonin(1A) receptors in juvenile Rhesus monkeys measured by positron emission tomography. *Biol Psychiatry* 2010; **67**: 1146–1153.
- 31 Vicentic A, Francis D, Moffett M, Lakatos A, Rogge G, Hubert GW et al. Maternal separation alters serotonergic transporter densities and serotonergic 1A receptors in rat brain. *Neuroscience* 2006; **140**: 355–365.
- 32 Burghy CA, Stodola DE, Ruttle PL, Molloy EK, Armstrong JM, Oler JA et al. Developmental pathways to amygdala-prefrontal function and internalizing symptoms in adolescence. *Nat Neurosci* 2012; **15**: 1736–1741.
- 33 Savitz J, Lucki I, Drevets WC. 5-HT<sub>1A</sub> receptor function in major depressive disorder. *Prog Neurobiol* 2009; **88**: 17–31.
- 34 Shrestha S, Hirvonen J, Hines CS, Henter ID, Svenningsson P, Pike VW et al. Serotonin-1A receptors in major depression quantified using PET: controversies, confounds, and recommendations. *Neuroimage* 2012; **59**: 3243–3251.
- 35 Hurley LM, Sullivan MR. From behavioral context to receptors: serotonergic modulatory pathways in the IC. *Front Neural Circuits* 2012; **6**: 58.
- 36 Kaneko WM, Riley EP, Ehlers CL. Effects of artificial rearing on electrophysiology and behavior in adult rats. *Depress Anxiety* 1996; **4**: 279–288.
- 37 Lesch KP, Waider J. Serotonin in the modulation of neural plasticity and networks: implications for neurodevelopmental disorders. *Neuron* 2012; **76**: 175–191.
- 38 Linnman C, Moulton EA, Barmettler G, Becerra L, Borsook D. Neuroimaging of the periaqueductal gray: state of the field. *Neuroimage* 2012; **60**: 505–522.
- 39 Bullmore E, Sporns O. Complex brain networks: graph theoretical analysis of structural and functional systems. *Nat Rev Neurosci* 2009; **10**: 186–198.
- 40 Jovanovic H, Perski A, Berglund H, Savic I. Chronic stress is linked to 5-HT<sub>1A</sub> receptor changes and functional disintegration of the limbic networks. *Neuroimage* 2011; **55**: 1178–1188.
- 41 Cisler JM, James GA, Tripathi S, Mletzko T, Heim C, Hu XP et al. Differential functional connectivity within an emotion regulation neural network among individuals resilient and susceptible to the depressogenic effects of early life stress. *Psychol Med* 2013; **43**: 507–518.
- 42 Gapp K, Jawaid A, Sarkies P, Bohacek J, Pelczar P, Prados J et al. Implication of sperm RNAs in transgenerational inheritance of the effects of early trauma in mice. *Nat Neurosci* 2014; **17**: 667–669.
- 43 Gapp K, Soldado-Magraner S, Alvarez-Sanchez M, Bohacek J, Vernaz G, Shu H et al. Early life stress in fathers improves behavioural flexibility in their offspring. *Nat Commun* 2014; **5**: 5466.
- 44 Elton A, Tripathi SP, Mletzko T, Young J, Cisler JM, James GA et al. Childhood maltreatment is associated with a sex-dependent functional reorganization of a brain inhibitory control network. *Hum Brain Mapp* 2013; **35**: 1654–1667.
- 45 Gould GG, Hensler JG, Burke TF, Benno RH, Onaivi ES, Daws LC. Density and function of central serotonin (5-HT) transporters, 5-HT<sub>1A</sub> and 5-HT<sub>2A</sub> receptors, and effects of their targeting on BTBR T+tf/J mouse social behavior. *J Neurochem* 2011; **116**: 291–303.
- 46 Hahn A, Stein P, Windischberger C, Weissenbacher A, Spindelegger C, Moser E et al. Reduced resting-state functional connectivity between amygdala and orbitofrontal cortex in social anxiety disorder. *Neuroimage* 2011; **56**: 881–889.
- 47 Stevenson RA. Using functional connectivity analyses to investigate the bases of autism spectrum disorders and other clinical populations. *J Neurosci* 2012; **32**: 17933–17934.
- 48 Finn ES, Shen X, Scheinost D, Rosenberg MD, Huang J, Chun MM et al. Functional connectome fingerprinting: identifying individuals using patterns of brain connectivity. *Nat Neurosci* 2015; **18**: 1664–1671.

Supplementary Information accompanies the paper on the Molecular Psychiatry website (<http://www.nature.com/mp>)